

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech.

Semester : VI

Branch :Electronics & Telecommunication Engineering and Allied

Subject Code & Name: Antennas and Wave Propagation [BTETC601]

Max Marks: 60

Date: 18/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)	CO01,02,03	12
1 Which of the following equations describes the power density of an electromagnetic wave in free space?		1
a. Friis Transmission Equation b. Maxwell's Equations		
c. Poynting Vector Equation d. Wave Equation		
2 Which of the following frequencies is typically used for ground wave propagation?		1
a. VHF b. UHF c. Microwave d. LF		
3 What is the MUF?		1
a. The highest frequency that can be used for sky wave propagation		
b. The lowest frequency that can be used for sky wave propagation		
c. The frequency at which the ionosphere is most reflective		
d. The frequency at which the ionosphere is least reflective		
4 Which of the following is NOT a type of antenna?		1
a. Dipole antenna b. Parabolic reflector		
c. Capacitor d. Horn antenna		
5 What is the main difference between a broadside array and an end-fire array?		1
a. The direction of maximum radiation		
b. The number of elements in the array		
c. The spacing between the elements		
d. The polarization of the elements		
6 Which of the following is a parasitic element in a Yagi-Uda antenna array?		1
a. The driven element b. The reflector		
c. The director d. Both reflector and director		

- | | | | |
|-------------|---|------------------------------|-----------|
| 7 | Which of the following is a common drawback of smart antenna systems? | | 1 |
| | a. High cost | b. Increased complexity | |
| | c. Reduced coverage area | d. All of the above | |
| 8 | What is the primary function of a horn antenna? | | 1 |
| | a. To enhance the directivity of the radiated wave | | |
| | b. To provide circular polarization | | |
| | c. To reduce ground losses | | |
| | d. To improve impedance matching | | |
| | What is the primary purpose of a lens antenna? | | |
| | a. To amplify the signal | | |
| | b. To focus the radiated energy | | |
| | c. To improve impedance matching | | |
| | d. To provide circular polarization | | |
| 10 | What is the characteristic radiation pattern of a small loop antenna? | | 1 |
| | a. Omni-directional | b. Figure-eight | |
| | c. Cardioid | d. Circular | |
| 11 | Which of the following is NOT a common application of smart antennas? | | 1 |
| | a. Cellular networks | b. Radar systems | |
| | c. Satellite communications | d. Power transmission | |
| | Antenna polarization refers to: | | |
| | a) The direction of the electric field of the radiated wave | | |
| | b) The direction of the magnetic field of the radiated wave | | |
| | c) The direction of the antenna's physical orientation | | |
| | d) The frequency of the radiated wave | | |
| Q. 2 | Solve the following. | | 12 |
| A. | Explain the concept of free space propagation and derive the Friis Transmission Equation. | CO 01 | 6 |
| B. | Differentiate between ground wave, sky wave, and space wave propagation. | CO 02 | 6 |
| Q. 3 | Solve the following. | | 12 |
| | Explain: | CO 02 | 6 |
| | (a) Radiation Pattern, | (b) Radiation Power Density, | |
| | (c) Radiation Intensity, | (d) Antenna Efficiency, | |
| | (e) Half-Power Beamwidth, | (f) Reciprocity Theorem | |
| B. | With the help of neat diagram explain radiation mechanism from two wire. | CO 02 | 6 |

Q. 4 Solve Any Two of the following.		12
A. Explain the two element uniform array and prove that	CO 02	6
$E = 2E_A \cos\left(\frac{\pi d \cos \theta}{\lambda} + \frac{\alpha_e}{2}\right)$		
B. Discuss the principle of pattern multiplication in an N-element linear array. How does the radiation pattern evolve as the number of elements increases?	CO 03	6
Compare the characteristics of a binomial and Dolph Tchebyshev antenna array. Explain the design principles behind each type and how they affect the antenna's performance.	CO 03	6
Q. 5 Solve Any Two of the following.		12
A. Compare adaptive and fixed beamforming techniques along with the distinguishing features of each, and in what scenarios are they most effective?	CO 02	6
B. Explain MUF. A high frequency radio link has to be established between two points as a distance of 2700 km on earth's surface. Consider the ionospheric height to be 250 km and its critical frequency 5 MHz, calculate the MUF for the given path	CO 02	6
C. Explain the advantages of smart antennas in wireless communication systems	CO 03	6
Q. 6 Solve Any Two of the following.		12
A. Explain travelling wave Antennas in detail.	CO 03	6
B. Explain the structural design and working principle of antennas with parabolic reflectors. Discuss the radiation pattern and the focusing properties of such antennas.	CO 03	6
C. Write a short note on (a) Rhombic Antenna and (b) V antenna.	CO 02,03	6

*** End ***